

Vishnu Sundhan

Software Engineer | GenAI & Agentic Systems | LLM Infrastructure
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Software Engineer with a background in Data Science who loves to learn and solve ambiguous challenges from scratch. I specialize in designing scalable microservices, Agentic systems, and data pipelines to build high-performance, reliable applications focused on the end user.

EDUCATION

New York University

M.S. in Computer Science | GPA: 3.78

New York, USA

Aug 2024 – May 2026

Rajalakshmi Engineering College, Anna University

B.E. in Computer Science and Engineering | GPA: 8.94/10

Chennai, India

Nov 2020 – May 2024

EXPERIENCE

Runova Inc

Software Engineer Intern

New York, USA

Jan 2026 – May 2026

- Developed a fully offline Android Skincare application in Kotlin by migrating a Flask REST API to native mobile, eliminated cloud dependency entirely, cut server-round-trip latency by 100%, and improved runtime stability and deployment reliability across distributed edge environments.
- Implemented LLM observability to track token cost and latency, used Veris AI's GCP-integrated simulation sandbox to stress-test and certify agents against accuracy, cost, and latency SLAs before customer deployment, partnering with stakeholders on go-live requirements.
- Engineered a real-time on-device processing pipeline using CameraX and Kotlin coroutines for facial analysis, designed an asynchronous parallel execution model supporting simultaneous computer vision inference, ML execution, and UI rendering within sub-100ms latency constraints.

Tata Technologies

Software Engineer Intern (Gen AI)

Pune, India

Jun 2025 – Jul 2025

- Designed and deployed multi-agent microservices using LangGraph, CrewAI, and FastAPI with a React/TypeScript frontend for context-aware project timeline coordination and a privacy-first cross-project meeting Assistant with ReAct-style reasoning for enterprises, reducing project ETA by 2 weeks.
- Fine-tuned PyTorch models for industrial machinery anomaly detection for pipe welding, achieving 96% precision and designed a live monitoring dashboard with optimized query caching that reduced diagnostic review time by 30%+, enabling faster fault identification on real time manufactured pipes.

Populus Empowerment Network

Software Developer Intern

Chennai, India

Jan 2024 – Jun 2024

- Built fault-tolerant ETL pipelines from ingesting various web sources and cleaning data to dashboard visualizations for processing 6M+ bilingual electoral records using Python, SQL, and Selenium. Maintained 98%+ data consistency and reliable downstream analytics workflows for stakeholders.
- Automated data extraction from ambiguous scanned documents using custom-trained CNNs, Coordinate Correction, Tesseract OCR, and Amazon Textract, which reduced manual processing effort by 50%+ and cut extraction error rates significantly across multiple document formats.

Rajalakshmi Engineering College

AI Research Assistant

Chennai, India

Oct 2023 – Jan 2024

- Published IEEE paper on deep learning-based Alzheimer's detection from MRI data, achieving 99.06% accuracy via a TensorFlow/PyTorch ensemble with custom OpenCV preprocessing and multi-stage feature extraction.
- Co-authored IEEE paper on brain MRI multi-class segmentation using CNN and Hugging Face transformer pipelines; validated classification on real-world clinical imaging datasets with high region detection accuracy.

PROJECTS

The Bitcoin Quorum – Autonomous Multisig AI Treasury

Apr 2026

LangGraph, LangChain, FastAPI, ChromaDB, Bitcoin Miniscript, Groq, Vapi, Next.js, Python

- Developed multi-agent autonomous Bitcoin treasury using LangGraph with 3-of-4 Miniscript multisig enforcement; agents independently think, vote, and cryptographically co-sign, making unilateral fund movement mathematically impossible – MIT Bitcoin Hackathon Community Pool Winner.
- Orchestrated parallel fraud detection across Tavily vendor search, Bitcoin Abuse blacklists, WHOIS domain screening, and Blockstream address analysis; integrated ChromaDB RAG for live policy retrieval and a Vapi voice interface for real-time treasury commands using Next.js as the interface.

Shadow Me - Privacy-Preserving Personal AI Agent

Mar 2026

LLMs, RAG, ReAct, Vertex AI, Presidio, BLIP, Whisper, ChromaDB, Telegram API, Python

- Built a conversational RAG AI agent with ReAct-style reasoning on 10K+ Telegram messages; integrated Microsoft Presidio to detect and redact 12 PII entity types before LLM inference, ensuring zero private data exposure prior to Vertex AI model serving.
- Engineered an offline multimodal pipeline processing images via BLIP captioning, OCR, and voice notes via Whisper transcription; converted all media to semantic embeddings for RAG retrieval, reducing raw data sent to the LLM by 90%+ versus direct transfer.

Refractor Radar – AI Code Analysis Agent

Nov 2025

TypeScript, Semantic Embeddings, AST Parsing, ChromaDB, CrewAI, NetworkX, Gemini API

- Built an AI-powered codebase analysis backend extracting AST structures and embedding functions via Gemini API; identified code duplication through semantic similarity clustering, enabling automated refactoring recommendations. Exposed the service via a Node.js web interface.

COMPETITIONS

3PercentHacks Hackathon – 1st Place

Jun 2025

Built *GameAble AI*, a hands-free gaming application using real-time pose estimation and a generative AI agent that dynamically creates custom pose-controlled game models with no additional hardware or peripherals.

ML Olympiad – Kaggle Competition 1st Place

Feb 2024

Achieved the lowest MAE of 1.33 among all participants by engineering novel features, building a stacked XGBoost/LightGBM ensemble, and applying systematic hyperparameter optimization; outperformed all teams on public and private leaderboards.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Kotlin, HTML/CSS

Backend & APIs: FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Node.js, React.js, LangGraph, CrewAI, LangChain, Google ADK, AutoGen, PydanticAI, Docker, Git, PyTorch Linux, CI/CD, Caching, System Design, Redis, Ollama

Frontend & Data: React, PowerBI, Pandas, NumPy, Matplotlib, Plotly, Jupyter, D3.js, GraphRAG, Hadoop, Apache Spark

ML & MLOps: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, Hugging Face, Transformers, MLflow, Prompt Engineering, Fine-Tuning, NLP

Cloud & Databases: GCP, Vertex AI, AWS SageMaker, AWS S3, DynamoDB, ChromaDB, Gemini API, Amazon Textract